



CISB3323 HUMAN COMPUTER INTERACTION

Final Report

Title of the Project: Math Explorer

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1.0 Introduction

1.1 Background

The Math Explorer project was inspired by the realization of setbacks faced by young learners using existing math educational applications. We have conducted ethnographic research with a popular application, where our study showed that though DoodleMaths gives practice on basic math exercises, it lacks offering adaptive learning pathways and personal feedback, the keystone of retaining engagement for long in young users.

Most of the math apps currently on the market suffer from non-intuitive interfaces, confusing ways of organizing difficulty levels using academic years instead of intuitive labels like "Easy," "Medium," and "Hard." This immediately creates a problem in that children cannot self-determine which level is most appropriate. Secondly, such apps often lack good gamification features with engaging progress tracking, thus not being well-suited for motivating users.

With Math Explorer, the ease of these challenges is a primary focus since the application aims at providing kids with a more friendly and interactive environment for learning. The application embeds personalized learning pathways, game-based rewards, and feedback mechanisms that would adapt to children's learning needs. With its interactive animations, voiceovers, and simplified interface, Math Explorer was designed to easily be used by children to enjoy their math learning process.

Our research has also indicated that children get tired quickly with the traditional exercises of mathematics, because they are repetitive and have static content. For this reason, in Math Explorer, the lessons are interactive, dynamic, and will move according to progress, thus encouraging self-confidence for long-term learning.

1.2 Ethnography

1.2.1 Observation

This consisted of three major tasks that needed to be performed by the users using the Math Explorer app during observation sessions, namely **Select the Level of Difficulty**, **Solve Mathematical Tasks**, and **Track Learning Progress**. These are common scenarios during typical use with a target application.

Task 1: Select the Level of Difficulty

The users were asked to work their way through the app and to select an appropriate difficulty level of their exercises. It was quite fascinating to see how most of the children, at first, did not understand the labeling of difficulties, especially when it was expressed in terms of 'year levels.' Most users had to be guided on which level to choose, and it showed that labeling should be simpler; for example, "Easy," "Medium," and "Hard," color-coded.

Task 2: Solve Mathematical Tasks

This exercise required users to interact and go through exercises on the mobile app, part of mathematics topics. We deduced that when little children face troubles in answering one question, usually they start randomly guessing answers thereafter. Animations and sound effects held our users' attention all along, even though some used to lose attention once the routines were repeated several times. The results indicated that quick feedback and hints increased motivation.

Task 3: Track Learning Progress

They were invited to log in to the progress tracker to view their achievements and rewards. While there was significant appeal with the visual feedback of progress, very few users actually used this feature to set goals or identify areas where they needed improvement. Instead, people were more interested in the progress tracking that was tied to the gamified rewards that encouraged them to continue using the app.

Overall, our findings pointed out that intuitive navigation, engaging content, and clear feedback mechanisms are the most important features in developing an interactive learning experience for young users.

1.2.2 User Analysis

Math Explorer targets primary school students, aged 6-10 years, where the development of foundational math skills is still at the initial stages. Users in this group are curious and motivated by interactive and visually appealing content. They have different levels of prior experience with educational applications; thus, intuitive navigation, personalized learning paths, and gamified rewards keep them interested and confident in their math skills.

WHO

Characteristics : Elementary school aged 6-10, intelligent and curious about math.

Skills : Not very much experience with mobile learning apps, not very much experience with touch navigation and teacher led interactions.

Education : Elementary School, still studying basics in math (addition, subtraction, multiplication, division).

Background : Probably had very little exposure to formal educational apps, most of their instruction will be supervised by an adult or parent.

WHAT

Goal : Engage, adaptive and confidence-building math with one-to-one lessons, interactive games, and gamified rewards.

Tasks : Select level of difficulty, solve mathematical tasks, track learning progress.

WHERE

Location : Home, school, or after school hours when users have a smartphone or tablet device in hand.

WHEN

Usage : Mostly in the class time, on homework, or in the free time for extra practice. Usage can be directed for instruction by parents/teachers.

HOW

System Access : Students access Math Explorer from their tablets or mobile devices. The app is touch friendly, has voiceovers, colorful animations, and it's gamified for easy use and fun.

1.2.3 Task Analysis

The **Hierarchical Task Analysis (HTA)** for the Math Explorer app highlights three core user tasks, broken down into actionable steps to ensure an intuitive and engaging user experience:

1. Select the Level of Difficulty:

Students can adjust the difficulty level of their activities through the app's settings. This task ensures the content is appropriately challenging, boosting confidence and encouraging skill development.

2. Solve Mathematical Tasks:

The app's primary feature guides students through interactive lessons, practice exercises, and personalized feedback. This task helps students learn math concepts, apply their knowledge, and improve through adaptive feedback mechanisms.

3. Track Learning Progress:

This task enables students to monitor their achievements, analyze performance, and identify areas for improvement. Through visual progress tracking and personalized recommendations, students can take control of their learning journey and set future goals.

These tasks, supported by the app's gamified elements and adaptive features, ensure a user-friendly and productive learning experience for young learners aged 6-10.

1- Select the Level of Difficulty

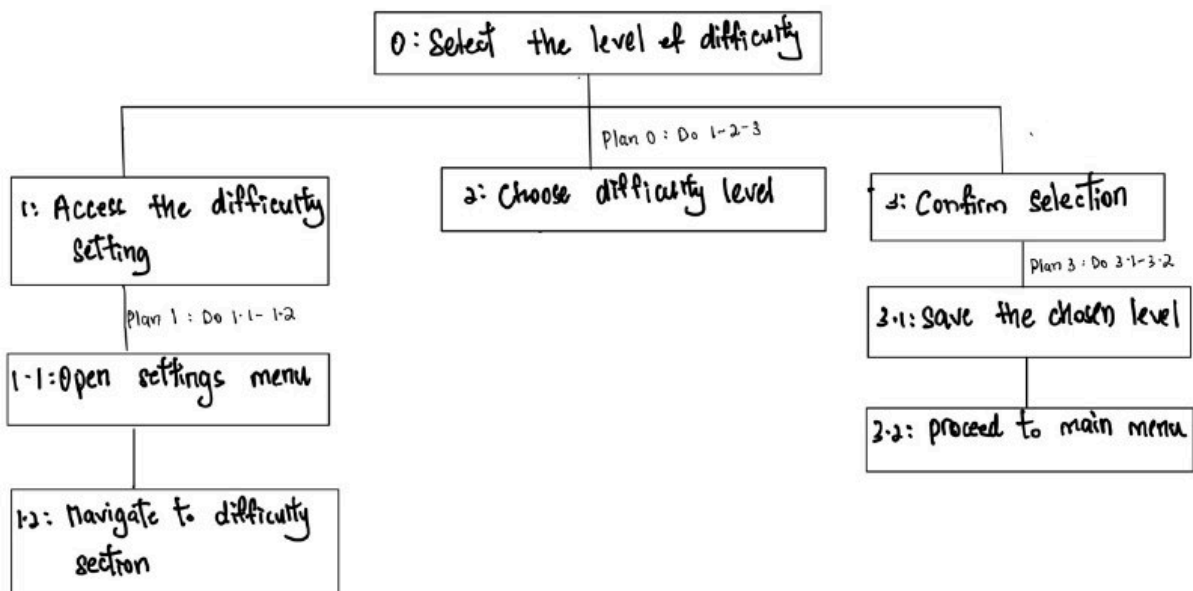


Figure 1.2.3.1 HTA-Main Task 1

2- Solve Mathematical Tasks

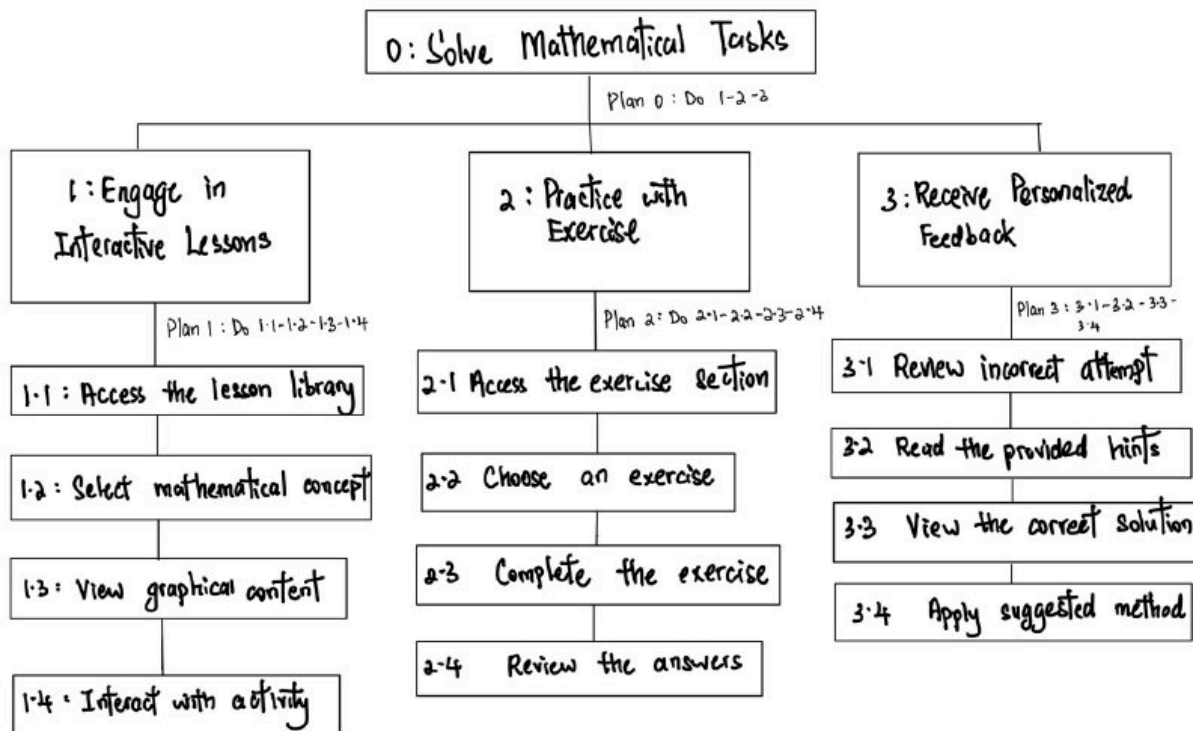


Figure 1.2.3.2 HTA-Main Task 2

3- Track Learning Progress

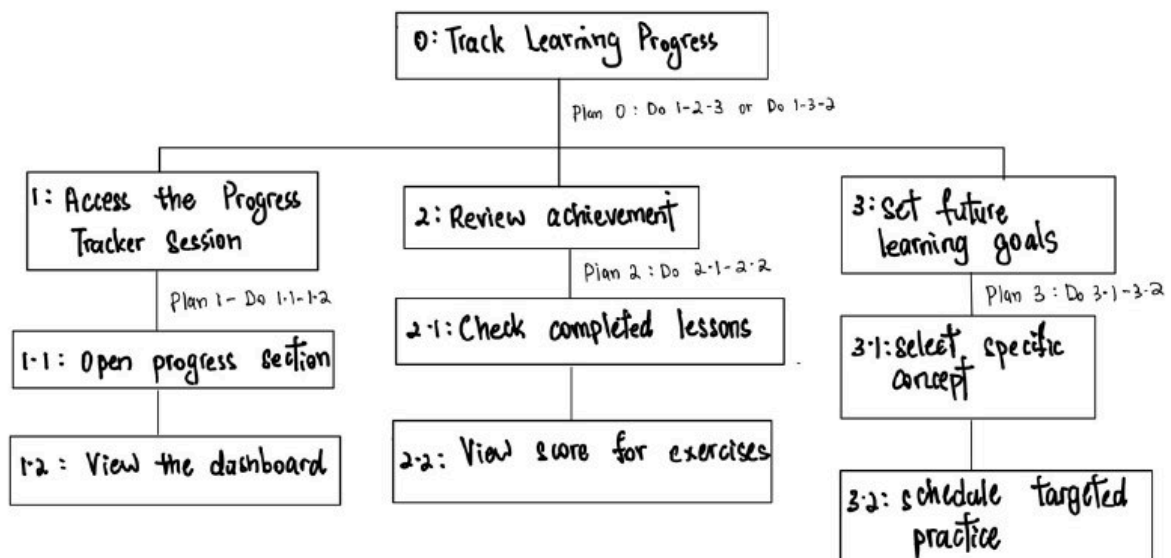


Figure 1.2.3.3 HTA-Main Task 3

1.2.4 Scenario

Sarah, an 8-year-old student at a primary school, had been so determined to excel in her math subject. Her mother lets her have the tablet in the living room, and she immediately opens her Math Explorer application and selects the multiplication lesson in an interactive way with colorful animations and voiceovers. After grasping the concept, she then proceeds to practice exercises, selecting the difficulty level at an intermediate. If she makes mistakes, it gives her instantaneous feedback and clues to improve. After the exercises are done, she goes to the dashboard, sees the progress, earns a badge, and notices personalized recommendations to revisit subtraction. Feeling accomplished and motivated, Sarah plans to do subtraction next, sure that finally she is able to advance in math.

2.0 Design

2.1 Design Techniques

2.1.1 Sketches

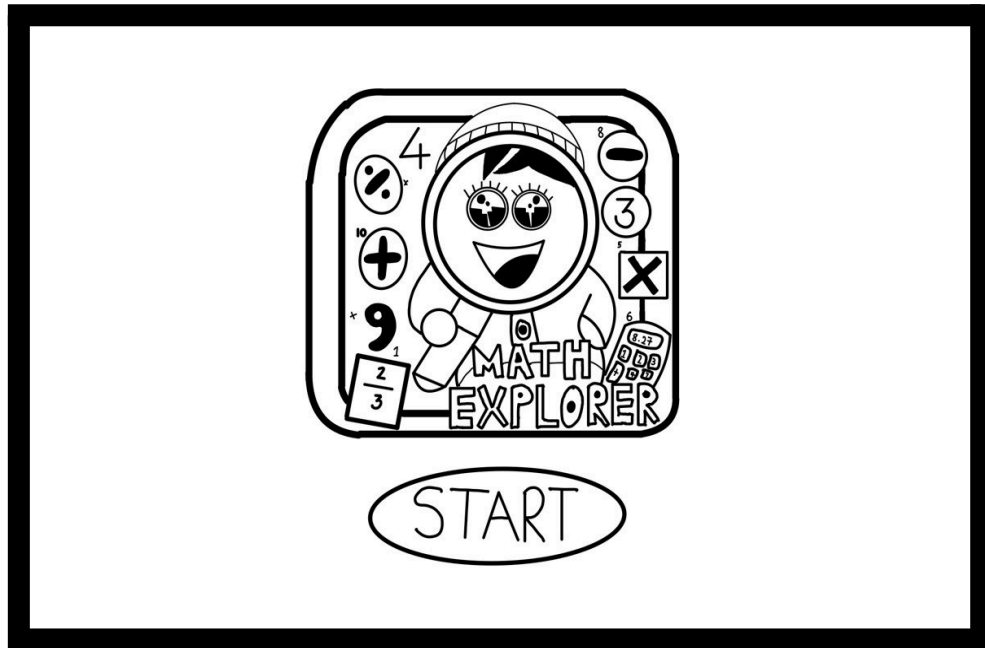


Figure 2.1.1 Application Logo

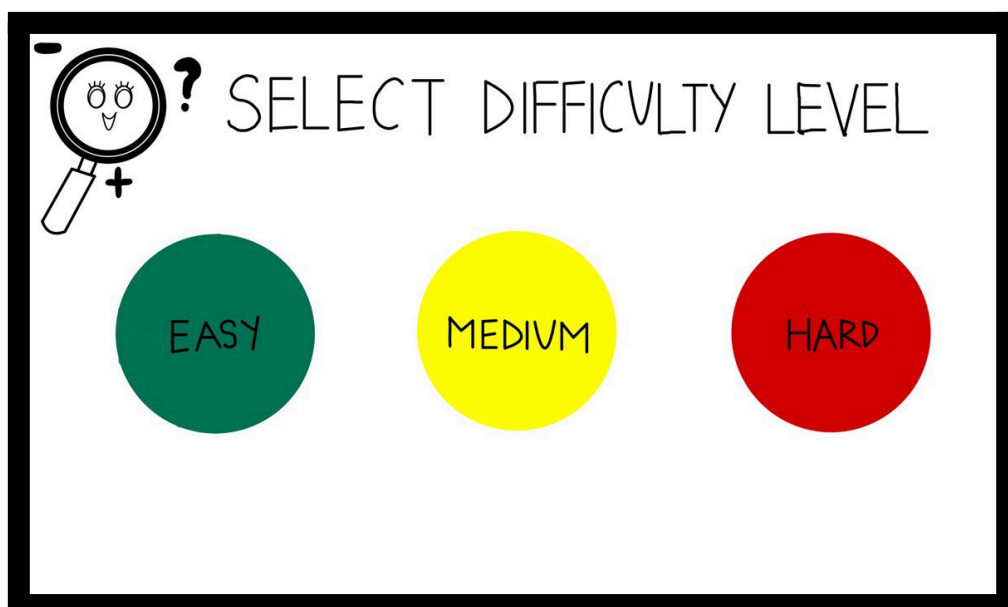


Figure 2.1.2 Main Task 1: Selecting difficulty level

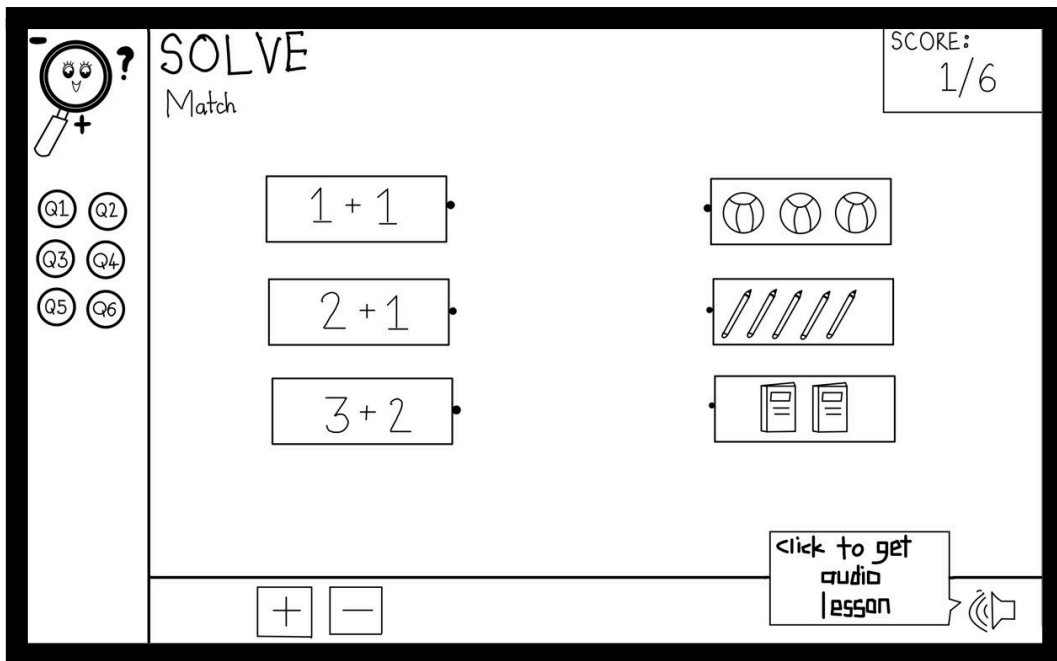


Figure 2.1.3 Main Task 2: Solving mathematical task

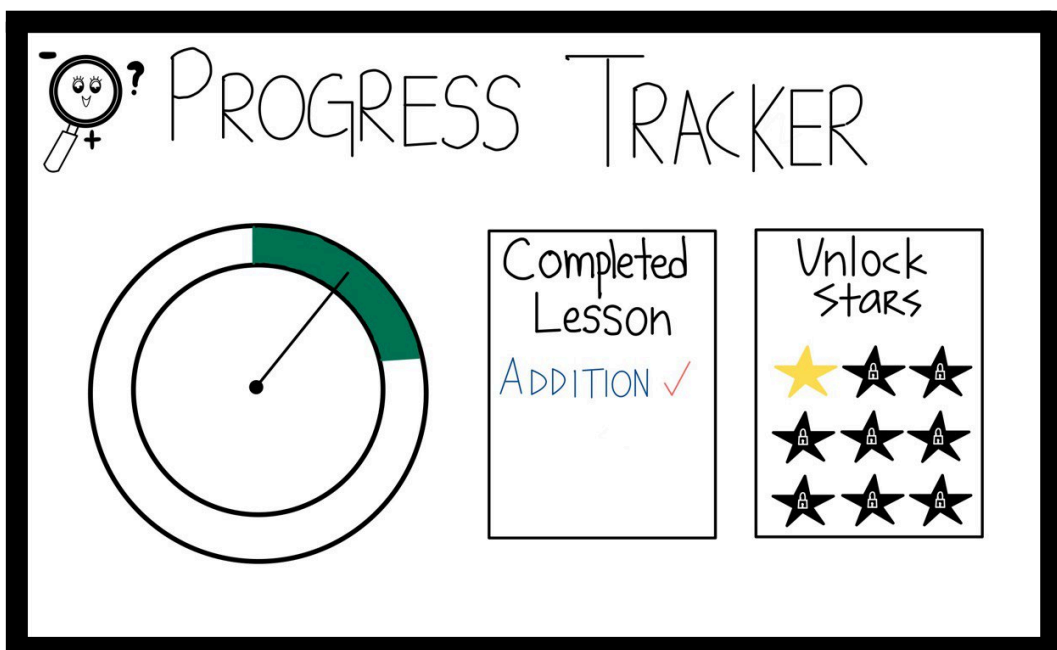


Figure 2.1.4 Main Task 3: Tracking progress after every completed lesson

2.2 Principle and Guidelines applied

2.2.1 *Schneiderman's 8 Golden Rules*

1. Strive for consistency
 - Uniform design elements like buttons, colors, and typography is used across all screens.
 - Green is used for easy mode, yellow is used for medium and red is used for hard mode.
 - A 'Next' button with consistent design is used in practice with exercise.
2. Enable frequent users to use shortcuts
 - Shortcuts or faster navigation is provided for experienced users.
 - 'Medal' icon in the top right corner of the Select Difficulty Level interface is a shortcut to progress tracker. Progress tracker can be accessed without going through practice with exercise interfaces.
3. Offer informative feedback for every user action
 - Clear and immediate responses are provided to user actions to enhance user confidence.
 - Correct answers were marked with a green tick and a pop up message with 'CORRECT', while incorrect ones showed a red cross and a pop up message with 'WRONG'.
 - Progress tracker after each lesson screen visually indicated the user's completion status.
4. Design dialogs to yield closure so that the users know when they have completed the process/steps
 - Users are made sure to feel a sense of accomplishment at the end of tasks.
 - After completing a lesson, the app displayed a "Lesson Completed" star along with an option to proceed to the next mode by having a 'Home' button on top right corner.
5. Offer error prevention and simple error handling
 - Errors are made easy to recognize and correct without frustration.
 - When choosing an answer, drag and drop is changed to clicking on the answers causing less error to be made by users.
6. Support internal locus of control
 - Users are empowered to feel in control of their interactions with the system.
 - Users could manually select their difficulty level for lessons and exercises.

7. Reduce short-term memory load
 - Interfaces are simplified to avoid overloading users with too much information at once.
 - Lessons are broken into bite-sized sections, with only a questions displayed per screen.
8. Minimize the user's cognitive load
 - The interface design is simple, making it easier for users to read and use the system efficiently.
 - Lessons are kept clear and straightforward, focusing on one task at a time.

2.3 Storyboard



Little girl feels sad for not being able to ace her Maths test.



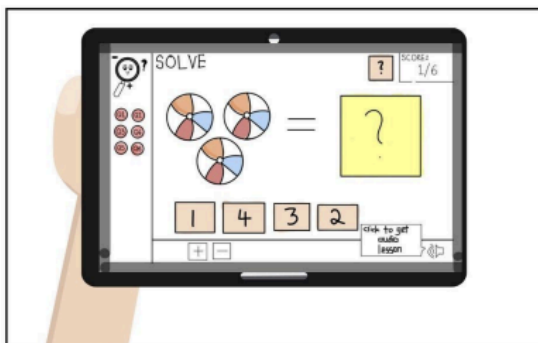
Her sister hands her the iPad.



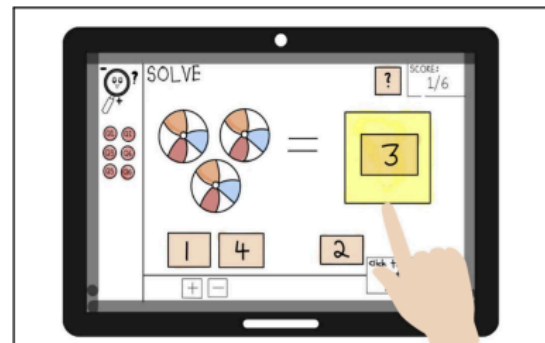
She opens the MathsExplorer application.



She chooses the difficulty level.



She tries answering the questions provided.



She selects the answer.

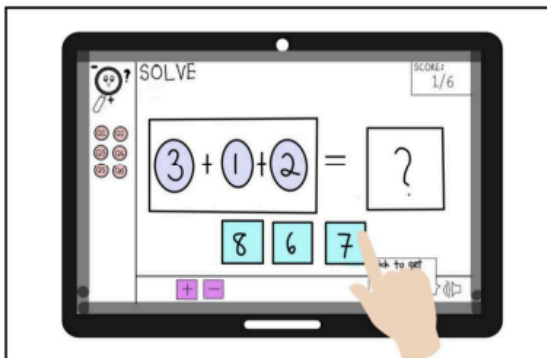
Figure 2.3.1 Storyboard 1



Correct display shows up.



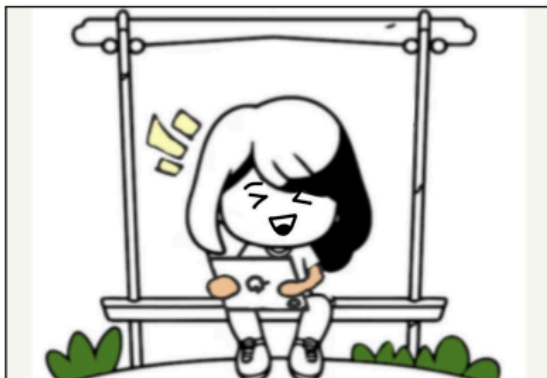
She cheers as she is happy that she got to answer it correctly.



She continues to solve until question 6.



The progress tracker shows the girl's development in the MathsExplorer app.



The girl feels excited as she got to unlock a star. She proceeds to do the subtraction exercise.

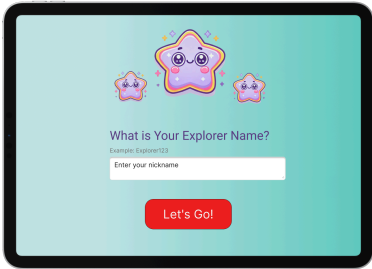
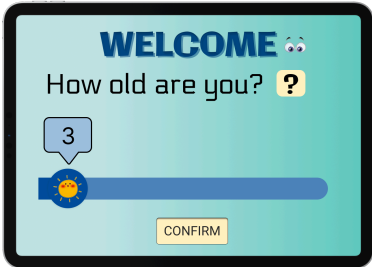


The girl gets to increase her rank and unlock more stars in the MathExplorer app.

Figure 2.3.2 Storyboard 2

3.0 Implementation

3.1 Implementation of the system

<i>Home Screen</i>	
 Figure 3.1.1: Welcoming Page 1 - START button	<u>Description:</u> First page of the system that includes a START button that initiates the interaction process.
	<u>Interaction:</u> Clicking the START button that navigates users to the <i>name input page</i>.
<i>Name Input Page</i>	
 Figure 3.1.2: Welcoming Page 2 - Input Name Section	<u>Description:</u> Allow users to input their name as part of setup progress.
	<u>Interaction:</u> Clicking the Let's Go button that navigates users to the <i>age selection page</i>.
<i>Age Selection Page</i>	
 Figure 3.1.3: Welcoming Page 3 - Select Age Section	<u>Description:</u> Allow users to select their age within the range between 3 to 10 years.
	<u>Interaction:</u> Drag the circle with the sun emoji to increase the age and click it to decrease. Users can hover the “?” to see a hint (Figure 3.1.4). By clicking the CONFIRM button to proceed to <i>main task 1</i>.

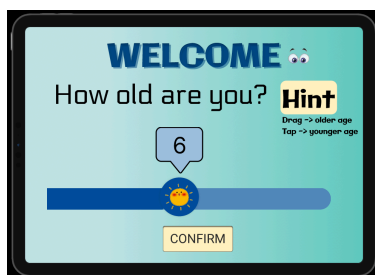


Figure 3.1.4: Hint message is shown when hovering

Main Task 1: Difficulty Selection Page

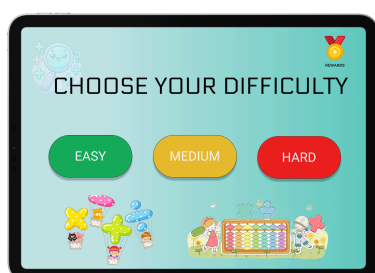


Figure 3.1.5: Main Task 1- Select the Level of Difficulty Section

Description:

The **reward** icon is positioned at the top right for clear visibility. Three different colors are used to represent the levels: green for easy, yellow for medium, and red for hard (Figure 3.1.5).

Interaction:

Click the **EASY** button to proceed with easy questions, the **MEDIUM** button to proceed with medium questions, and the **HARD** button to proceed with hard questions.

Main Task 2: Solve Mathematical Tasks Page



Figure 3.1.6: Main Task 2 - Interactive Section - EASY MODE

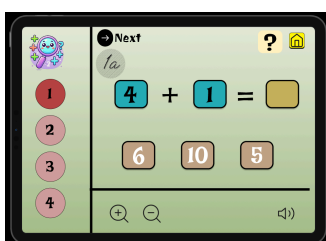


Figure 3.1.7: Main Task 2 - Practice with Exercises Section - EASY MODE - Question 1a

Description:

1. Interactive Section (Figure 3.1.6)

The **home** icon is placed at the top right, with the **profile** icon next to it. The **EXERCISE** button is located below them. The video thumbnail is on the left, and a bottom navigation bar is positioned beneath the thumbnail. **Tips** are also provided, placed beside the video.

2. Practice with Exercises Section (Figure 3.1.7)

The question number is displayed on the left and changes color to indicate the current question (Figure 3.1.7 & Figure 3.1.8). The **Home icon** is placed at the top right, while the **Next** button is

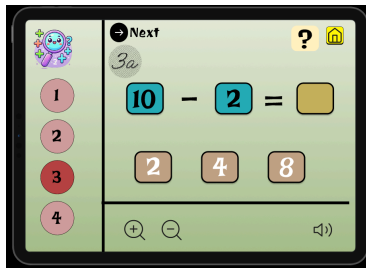


Figure 3.1.8: Main Task 2 - Practice with Exercises Section - EASY MODE - Question 3a

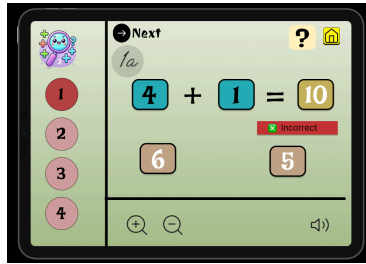


Figure 3.1.9: Incorrect Symbol is shown if choose the wrong answer

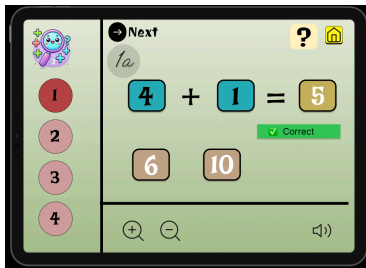


Figure 3.1.10: Correct Symbol is shown if choose the right answer

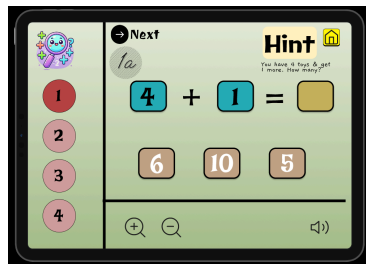


Figure 3.1.11: Hint message is shown when tap the ? icon

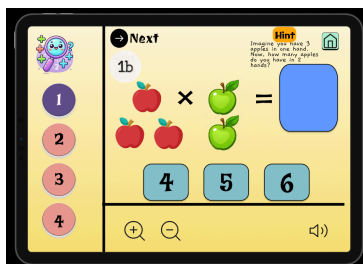


Figure 3.1.12: Main Task 2 - Practice with Exercises Section - MEDIUM MODE - Question 1b

at the top left. The mathematical task is centered on the page for easy focus. The “?” icon provides a hint to help users solve the task. Different background colors are used to indicate the levels of difficulty.

Interaction:

1. Interactive Section (Figure 3.1.6)

Users can quit the lesson at any time by clicking the **Home icon** to return to the *difficulty selection page*. By clicking the **EXERCISE** button, users can navigate to the exercise section.

2. Practice with Exercises Section (Figure 3.1.7)

By clicking the question number, users can navigate to another question. They can select from the options provided to answer the question. If the answer is correct, a correct symbol will appear (Figure 3.1.10) ; if not, an incorrect symbol will be shown (Figure 3.1.9). Users can quit the question at any time by clicking the **Home icon** to return to the *difficulty selection page*. If users face any problem, they can click the “?” icon to show the hint.

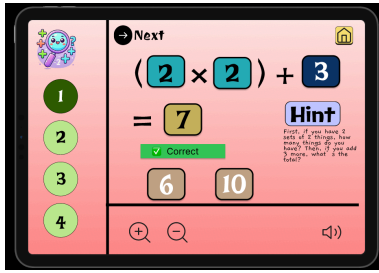


Figure 3.1.13: Main Task 2 - Practice with Exercises
Section - HARD MODE - Question 1

Main Task 3: Learning Progress Tracking Page

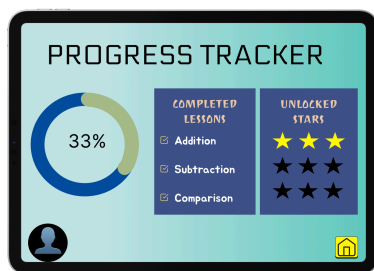


Figure 3.1.14: Main Task 3 - Track Learning
Progress Section 1

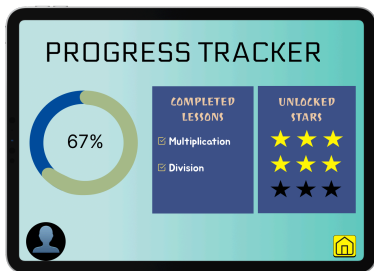


Figure 3.1.15: Main Task 3 - Track Learning
Progress Section 2

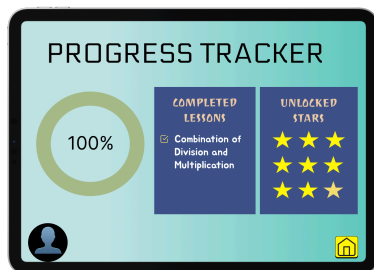


Figure 3.1.16: Main Task 3 - Track Learning
Progress Section 3

Description:

The percentage and circle tracker are located on the left. The completed lessons and unlocked stars are positioned next to the percentage. The **home icon** is at the bottom right, while the **profile icon** is at the bottom left.

Interaction:

After completing the easy mode questions, users will be navigated to the progress tracker, showing that they are **33%** done (Figure 3.1.14). Upon finishing the medium mode questions, the tracker will update to **67%** (Figure 3.1.15), and after completing the hard mode questions, it will show **100%** (Figure 3.1.16). The completed lesson section will display the topics covered. For example, addition, subtraction, and comparison for easy mode. Each completed task will be ticked off, and unlocked stars will be shown to indicate progress. Users can click the **home icon** to return to the *difficulty selection page*.

3.2 Testing/Evaluation plan

3.2.1 Objective

To evaluate both the usability and functionality of the Math Explorer app, ensuring it is user-friendly and performs as expected for children aged 6–10 years.

3.2.2 Methodology

Usability Testing

Participants: 5 target users (children aged 6-10 years) were invited to interact with Math Explorer and complete the main tasks.

Main Tasks

Main Task 1: Select the Level of Difficulty

Users were required to select one of the three modes: easy, medium, or hard.

Main Task 2: Solve Mathematical Tasks

Users answered math questions from the selected difficulty mode to test the question interface and answer selection process.

Main Task 3: Track Learning Progress

After completing the math tasks, users reviewed their progress and unlocked stars, which were shown immediately after finishing each mode.

Functional Testing

The app was tested to ensure it performs as expected, including intuitive navigation and error-free operations.

3.2.3 Metrics

Main Task	Observation	User's Feedback	Recommendation for Improvements
Select the Level of Difficulty	All users successfully selected a difficulty level	Question: How would you rate the difficulty selection interface? (1: Very Unsatisfied – 5: Very Satisfied) Answer: 5/5	- None needed.
	Average completion time: 15 seconds.		
	No errors or confusion encountered.		

Solve Mathematical Tasks	Average time spent: Easy mode - 2 minutes per question Medium mode - 2.5 minutes per question Hard mode - 3 minutes per question	Question: How would you rate the interaction for solving tasks? (1: Very Difficult – 5: Very Easy) Answer: 4/5	- Provide the correct answer after submission or as part of a feedback feature to support learning. - Move the "next" button to a bottom center to prevent confusion.
	Tap-and-select interaction worked smoothly with no errors or confusion.	Question: What did you find most useful about the question interface? Answer: Hint is provided for every question.	
	Users appreciated the straightforward interface and ability to confirm their answers.	Question: What problem did you face when going through the question? Answer: - Cannot undo when selecting the wrong answer. - The correct answer is not provided. - Confused about going to the next page, as the next button is placed at the top left.	
Track Learning Progress	All users accessed the progress report without major issues as it was shown immediately after completing each mode.	Question: How would you rate the progress report feature? (1: Very Unsatisfied – 5: Very Satisfied) Answer: 4.5/5	- Update the interface with brighter and more to make it visually appealing for children, as the current design appears dull. - Provide review materials for topics where the user consistently makes mistakes.
	Average time spent reviewing progress: 45 seconds.	Question: Did the progress report motivate you to keep improving? (Yes/No) Answer: Yes as I want to unlock all the stars.	
	Users liked the clear display of completed tasks and unlocked stars.	Question: What additional information would you like to see in your progress report?	

		Answer: I would like to see any areas I need to focus on, such as multiplication, as I always choose the wrong answer.	
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4.0 Conclusion

4.1 Conclusion of the project

In conclusion, this project centers on designing a mobile learning app that is both practical and engaging for primary school students. Through sketching and storyboarding, we have outlined the app's core functionalities while adhering to key design principles such as usability, consistency, and accessibility.

The app aims to enhance students' mathematical skills by offering interactive lessons, personalized feedback, and progress tracking. Its straightforward and child-friendly design ensures ease of use, minimizing potential challenges for young learners.

This project highlights the value of applying HCI principles to create user-centered solutions. As the design progresses, further refinement and usability testing will be essential to ensure the app meets its objectives effectively and provides a meaningful learning experience.

4.2 Limitation

1. **Inability to Edit Answers:**

One significant limitation of the current design is the inability for users to revisit and modify their previous answers. Once an answer is submitted, users are unable to go back to make changes, which could lead to frustration if they realize they've made a mistake or wish to provide a different response. This limitation compromises user flexibility and may negatively impact the overall user experience, especially in scenarios where accuracy is critical or when users are unsure about their selections.

2. **'Next' Button Placement:**

The positioning of the 'Next' button on the interface is another limitation. Currently placed in a location that may not be immediately accessible or intuitive for all users that can result in inefficient navigation. Users might need to search for the button, leading to unnecessary cognitive load and delays in completing tasks. A more user-friendly approach would be to position the 'Next' button at the lower-right corner, as this is a common convention in many applications, making it easier for users to navigate through the interface quickly and smoothly.

3. **Age Selection Usability:**

The age selection feature is currently not user-friendly, particularly for our target user group. The method of selecting age might involve excessive dragging or complex interactions, which could confuse users or make it harder for them to input their age efficiently. For younger or less tech-savvy users, this could lead to frustration and a less enjoyable user experience.

4.3 Future Enhancement

1. **Allow Editing of Answers:**

Incorporating a feature that allows users to revisit and modify their answers could enhance flexibility and user satisfaction. This can be implemented by adding a "Back" button that enables users to go back to previous questions without losing the progress they've made. Alternatively, providing a review screen at the end where users can edit their responses before final submission would also improve user control. Additionally, implementing a confirmation prompt before final submission could help avoid errors and improve the overall experience.

2. **Improve 'Next' Button Accessibility:**

To make the 'Next' button more accessible, consider placing it consistently in the lower-right corner, with larger size and higher contrast to make it easier to find, especially for users with visual impairments. In mobile or tablet designs, consider ensuring the button is reachable for users using one hand, which may involve testing its position across different screen sizes. Another potential enhancement would be incorporating a "keyboard shortcut" for quick navigation through questions or a visual progress indicator to reduce cognitive load and allow users to track where they are in the process.

3. **Revise Age Selection Interface:**

To improve the age selection experience, consider offering a more intuitive input method, such as a **drop-down list**, which would make selecting an age faster and more user-friendly. The drop-down list could have preset values, making it easy for users to pick an age quickly. For users who are less familiar with such interfaces, providing tooltips or instructional hints explaining the method can further improve usability. Additionally, for younger audiences, implementing an age verification system that is both simple and engaging, such as a quiz or age-related avatars, could increase engagement and ensure a seamless experience.

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Prototype:

Figma. (2024). Figma.

<https://www.figma.com/proto/07gdJw1OG9C5xDNuhQIgl/Math-Explorer?node-id=1-3484&p=f&t=4qQQTi5a3B4c7VTS-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A3484&show-proto-sidebar=1>

6.0 Appendix

6.1 Questionnaire used for interview

1. How easy was it to navigate the app? (1: Very Difficult – 5: Very Easy)
2. Were the instructions for each task clear and understandable? (Yes/No)
3. Did you encounter any difficulties while completing tasks? (Yes/No, please specify)
4. Did the buttons (Next, Home) work as expected? (Yes/No)
5. Were the symbols (correct/incorrect) displayed accurately after answering a question? (Yes/No)
6. Did the app transition smoothly between pages? (Yes/No)
7. How enjoyable was your experience with the app? (1: Not Enjoyable – 5: Very Enjoyable)
8. Would you recommend this app to others? (Yes/No)
9. Any additional feedback or suggestions for improvement?